

Towards SOP-based Real-Time Earthquake Detection: Limitations and Opportunities

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Abstract We establish detectability limits of SOP-based earthquake sensing using a physics-informed statistical labeling model on real data: 95% of 798 events have SNR below 1.64 dB, independent of distance. We demonstrate a real-time, low-cost seismic event pipeline supporting up to 20 Hz Jones-matrix sampling. ©2026 The Author(s)

Introduction

Submarine coherent optical cables form a dense, always-on sensing network across the ocean floor. By analysing state-of-polarisation (SOP) data already available in commercial coherent transceivers, earthquake-induced disturbances can be identified without dedicated seafloor sensors [1]. The principle is established: seismic strain changes fibre birefringence and rotates the optical polarisation, while the transceiver digital signal processing (DSP) continuously tracks this rotation during normal operation [2]. Several studies have demonstrated SOP-based earthquake identification on transoceanic cables [3–5].

Despite these results, key challenges remain before practical deployment. First, polarisation measurements are cumulative along the fibre, so perturbations from different sections combine, making localisation and interpretation difficult [2]. Second, prior studies mainly considered a small number

of strong earthquakes, leaving detectability under realistic conditions unclear. Third, large-scale deployment requires continuous real-time detection on multiple transponders using low-cost embedded hardware with a balance between accuracy and false alarms. Therefore, a clear detectability baseline is still missing: which earthquakes can be reliably detected, at what magnitudes and distances, and with what performance using low-complexity methods. Without this baseline, the operational value of SOP sensing remains uncertain.

This article addresses this gap through the first broad statistical study of SOP-based earthquake detection on a deployed coherent link, combined with benchmarking of embedded real-time detectors. We show that most seismic events produce SOP signals comparable to background noise, fundamentally limiting detectability. We analyse 798 seismic events measured from commercial

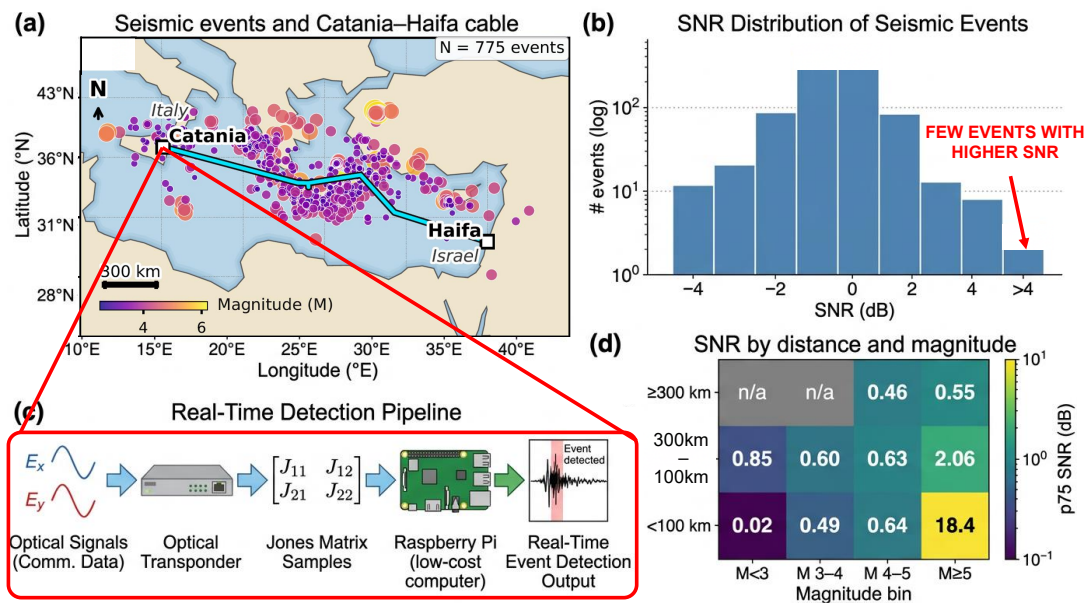


Fig. 1: (a) Set of real seismic events used for statistical analysis. (b) Event SNR distribution demonstrating that the majority of events that reach the cable have a poor SNR. (c) Heat map to show that, also among the event close to the cable, only those with high magnitude presented a high SNR.

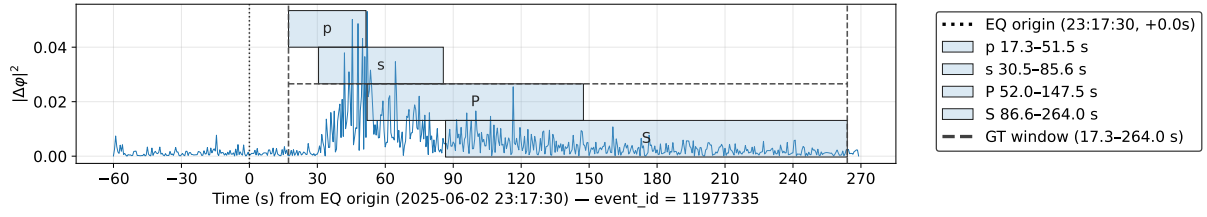


Fig. 2: Theoretical event-arrival estimate (Ground Truth-GT) for earthquake 11977335 on the Catania–Haifa cable (MedNautilus system). Blue bands show the predicted “p”, “s”, “P”, and “S” phase windows (earliest to latest arrival along the cable). The dashed box marks the GT window, from earliest “p” to latest “S”. Waveform shows the squared rotation-vector magnitude ($|\Delta\varphi|^2$).

transponders as the one presented in [6] and observe that 95% have SNR below 1.64 dB. Even events close to the cable often require magnitude above 5 to produce strong event signals. We evaluate low-complexity moving-average and P95 threshold detectors, reporting their true-positive and false-positive trade-offs as a practical baseline. We then implement a real-time pipeline on low-cost hardware using real data, supporting Jones-matrix sampling up to 20 Hz. High-SNR events are detected with up to $10\times$ lower latency, while meaningful accuracy with low false alarms requires SNR above 4 dB. Finally, we show that the rotation-vector method of Mecozzi [1] requires $\approx 10\times$ more resources than the proposed pipeline. These results provide a practical baseline for deploying SOP-based sensing in operational submarine communication networks. Overall, low-SNR event detection remains the main challenge to unlock submarine cables as a global seismic sensing network.

Real-World Seismic Events Statistics: Figure 1 summarizes the signal-to-noise ratio (SNR) statistics from State of Polarization (SOP) measurements on a Mediterranean submarine cable, using the dataset of [7]. Following [1], the measured Jones matrices are converted into a rotation vector, used here as the sensing variable. The SNR is defined as $\text{SNR} = 10 \log_{10}(P_{\text{signal}}/P_{\text{noise}})$, where P_{signal} is the event power and P_{noise} is the pre-event background power. Fig. (a) shows the locations of the $N = 798$ earthquakes analysed, fetched from public earthquake catalogs (e.g. IRIS), covering local, regional, and teleseismic distances relative to the cable route (light blue line). Fig. (b) shows the SNR distribution, approximately symmetric around 0 dB, with most

events within $[-2, +2]$ dB. Thus, for most earthquakes, the SOP variation is similar to normal background fluctuations, making detection difficult. Fig. (c) illustrates a representative output on field data: two co-seismic arrivals are clearly visible in the filtered amplitude (top), produce threshold crossings exceeding the decision level (middle), and are correctly declared as events (bottom) with no false alarms in the displayed window. Fig. (d) shows mean SNR versus distance and magnitude ($M < 3$, $M 3-4$, $M 4-5$, $M \geq 5$). For local events, mean SNR rises with magnitude, from 0.019 dB for $M < 3$ ($n = 31$) to 18.4 dB for $M \geq 5$ ($n = 6$). This is the only group where the signal clearly exceeds the background.

For regional and teleseismic events, mean SNR stays near or below 1 dB in all magnitude ranges, except for the four largest regional events ($M \geq 5$), which reach $\text{SNR} = 2.06$ dB.

Overall, reliable SOP-based earthquake detection on this cable is mainly limited to large events close to the cable. SNR drops quickly with increasing distance or decreasing magnitude.

Physics-Informed Model for Event Labeling: for each earthquake, we estimated the arrival times of the main body-wave phases (p, P, s, S) using ObsPy’s `TauP` module [8, 9] with the AK135 Earth model [10].

Because the cable spans $\approx 2,000$ km, seismic waves reach different cable sections at different times. To capture this, arrivals were calculated at multiple points along the cable, spaced every 6 km. For each phase, we then defined a time window between its earliest and latest predicted arrival across the cable.

Some source–receiver geometries do not produce all phases (for example p or s); unavailable

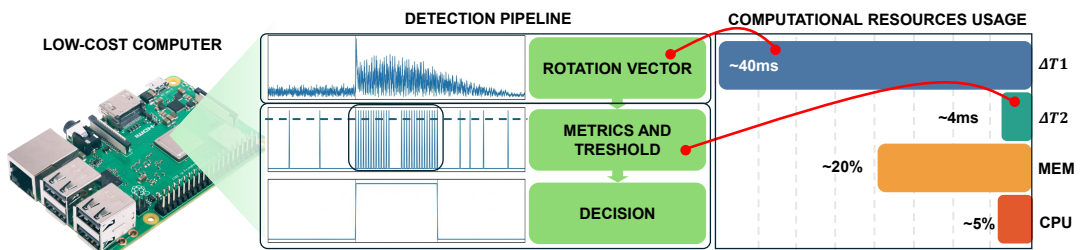


Fig. 3: Real-time embedded pipeline performance. Total processing time is ≈ 50 ms ($\Delta T_1 + \Delta T_2$), enabling up to 20 Hz Jones-matrix sampling. The rotation-vector stage dominates cost ($\Delta T_1 \approx 40$ ms), while metrics-and-threshold requires $\Delta T_2 \approx 4$ ms ($\sim 10\times$ faster). Full system usage is $\approx 5\%$ CPU and 20% memory.

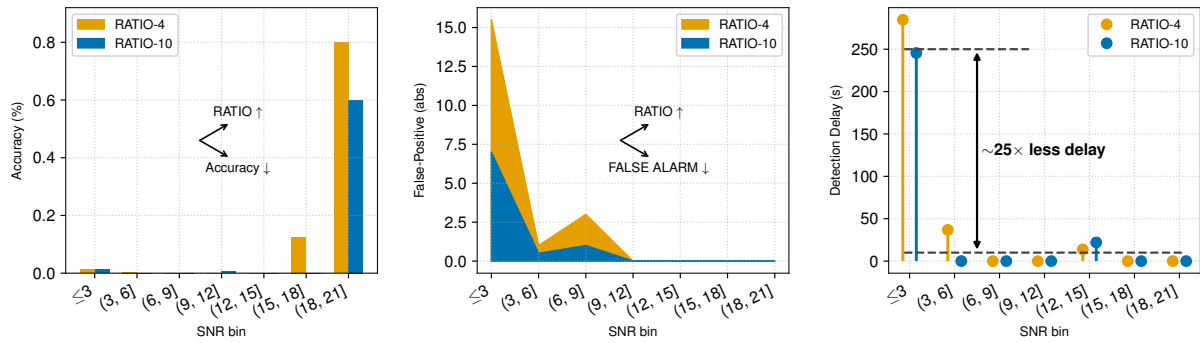


Fig. 4: a) and b) show the threshold-ratio trade-off: higher ratios reduce false alarms but also lower detection accuracy. In both cases, higher SNR improves accuracy and reduces false alarms. c) shows that detection latency is independent of ratio and is mainly determined by SNR.

phases were ignored. The Ground Truth (GT) event window was defined as the overall envelope of all available phases, from the earliest predicted arrival to the latest one (Fig. 2). Therefore, the labeled event window contains the full time interval where modeled seismic body-wave energy is expected to be present in the SOP measurements.

Low-cost Real-Time Detection Pipeline

Figure 3 summarises the real-time implementation of the detection pipeline and its computational cost on a low-cost single-board computer, demonstrating feasibility for deployment on low-cost edge devices. The pipeline receives the Jones-matrix stream from the coherent transceiver and processes it in three stages: (i) a *rotation-vector* stage that converts polarisation samples into a seismic-sensitive time series, (ii) a *metrics-and-threshold* stage that computes a moving-average or P95 detector ratio between the event window and pre-origin window and applies a fixed ratio threshold, and (iii) a *decision* stage that outputs a binary event flag. The waveforms show the signal evolution from seismic activity at the rotation-vector output to a clean square-wave decision signal.

The measured execution time and resource usage reveal that the system meets real-time requirements, requiring a minimum of ≈ 50 ms ($\Delta T_1 + \Delta T_2$), resulting in a maximum 20Hz Jones-matrix sampling rate. The rotation-vector stage is the main cost, requiring $\Delta T_1 \approx 40$ ms per window, while the metrics-and-threshold stage requires only $\Delta T_2 \approx 4$ ms, about $10\times$ faster. The full pipeline uses about 5% CPU and 20% memory on the target platform, leaving headroom for other monitoring tasks.

These results quantify the accuracy-versus-cost trade-off: the rotation-vector method of Mecozzi [1] is effective but around $10\times$ more expensive than the statistical detection stages, motivating selective use in resource-constrained systems.

Detection Limitations and Opportunities

Figure 4 shows detection performance versus SNR for two threshold ratios (RATIO-4 and RATIO-

10) and two metrics: moving average (MA) and 95th percentile (P95). It demonstrates that detection performance is primarily limited by SNR rather than threshold selection.

Panels (a) and (b) show the threshold trade-off. Increasing the ratio from 4 to 10 reduces false alarms by about $2\times$ in the low-SNR region ($\text{SNR} \leq 3$ dB), but also lowers true detections. In the higher $\text{SNR} \geq 4$ dB group, the true-positive rate falls from $\sim 80\%$ to $\sim 60\%$. Both metrics improve steadily with SNR, showing that background noise, more than threshold choice, is the main limit to performance. Panel (c) shows that detection latency depends mainly on SNR and is almost unaffected by threshold ratio. Events with $\text{SNR} \leq 3$ dB have median latency of ~ 250 s, while events with $\text{SNR} \geq 6$ dB are detected in ~ 10 s, an improvement of about $25\times$.

These results have two implications. First, within simple threshold methods, no single ratio can simultaneously minimise false alarms and maximise detection in low-SNR conditions. Second, the strong dependence on SNR shows that low-SNR detection is the main challenge. Future methods should better exploit signal structure, such as spatio-temporal cable coherence, matched filtering, or data-driven models, to improve sensitivity while keeping low false alarms.

Conclusions

Applying statistical analysis to 798 seismic events we identified that 95% have SNR below 1.64 dB, meaning reliable detection is mainly limited to large nearby earthquakes. Low-complexity moving-average and P95 detectors has a clear trade-off between false alarms and accuracy, with useful performance requiring SNR above 4 dB. This implies that only a small subset of events can be reliably detected using simple threshold-based methods. A real-time embedded implementation confirmed practical operation, while the rotation-vector stage dominated computational cost. These results highlight that improving low-SNR detection is the key requirement for large-scale deployment of SOP-based sensing.

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